Supplementary Information for
A positive-electrode model separating iodine saturation from accessibility loss in zinc–iodine flow batteries

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Contents

This Supplementary Information reports source identity, equations, numerical checks, complete field inventories and bounded lower-scale calculations for the porous ZIFB positive electrode. Existing full-cell records are descriptive anchors; molecular and mesoscale calculations are priors or comparators. Neither evidence class identifies deposit morphology or independently validates an internal positive-electrode state.

Evidence classes are used consistently below. `E-EXP` denotes registered cell-level reanalysis; `E-SIM`, a registered continuum solution or exported field; `E-CTRL`, a matched model-form control; `E-COMP`, a deterministic lower-scale calculation; `E-POST`, a postprocessed scenario; and `E-LIT/PRES`, a literature scale or prescribed input. These labels describe permitted inference, not a ranking across unlike evidence types.

S1 Existing full-cell records and data provenance

S1.1 Cell formats, source identity and inclusion rules

Galvanostatic capacity, composition, compression and rate records were acquired in $4\,\mathrm{cm}^2$ full cells with a porous iodine positive electrode, zinc negative electrode and Nafion 212 separator. Separate cyclic- voltammetry, chronoamperometry and impedance records used a $1\,\mathrm{cm}^2$ three-electrode configuration with an Ag/AgCl reference; no full-cell capacity is described as a three-electrode measurement. The physical cell/file is the independent unit, and sequential cycles are repeated observations within that unit. The retained acquisition records do not contain an independently logged experimental temperature. Continuum and MD temperatures are therefore not assigned retrospectively to the experiments. Under correction ID `EXP-META-001`, every legacy acquisition filename or condition field containing `NH4Cl`, including capitalization variants, denotes an experiment performed with NH4Br. Raw names, bytes and hashes remain unchanged; derived labels use the corrected identity.

Table S1: Experimental datasets, physical units and analysis roles. Each row identifies the physical cell or file unit, protocol role, processed source and permitted descriptive comparison. Dataset rows can overlap in source cells and are not summed as an experiment-wide sample size.

dataset	cell format	physical cells	cycles/files	processed source	evidence boundary
late-charge derivative	4 cm^2 full cell	1	4 cycles	`battery_experiment/02_processed_data/R581_G4_DVDQ_REBUILD`	within-cell estimator sensitivity only
composition series	4 cm^2 full cell	6	8 included run files; 315 classified cycles	`battery_experiment/02_processed_data/R581_COMPOSITION_AXIS_REBUILD`	descriptive condition-associated envelopes; no population law
compression series	4 cm^2 full cell	4	108 of 110 reconstructed paired cycles retained	`battery_experiment/02_processed_data/R581_COMPRESSION_EVIDENCE_REBUILD`	protocol brackets; one cell per thickness
pristine rate ladder	4 cm^2 full cell	1	29 cycles	`battery_experiment/02_processed_data/R581_4M_RATE_HIGH_LOADING_REBUILD`	same-cell sequence at 80 mA cm^{-2} to 360 mA cm^{-2}
cross-protocol marker	4 cm^2 full cell	1	4 cycles	same processed directory	separate 40 mA cm^{-2} cell; not connected to the rate ladder
high-rate material proxy	4 cm^2 full cell	1	4 cycles	same processed directory	400 mA cm^{-2} P350C-positive proxy; not a pristine continuation
electrochemical context	1 cm^2 three-electrode cell	not pooled in R582	separate CV/CA/EIS files	`electrochemical`	method context only; not full-cell capacity evidence

S1.2 Sensitivity of the late-charge voltage feature

The derivative analysis used the full-resolution pristine source, retained all four eligible sequential cycles and interpolated voltage to a 0.25 mAh cm^{-2} grid. A cubic Savitzky–Golay derivative [1] with a 10.25 mAh cm^{-2} physical window was the primary estimator; 5.25 mAh cm^{-2} and 15.25 mAh cm^{-2} were prespecified sensitivity windows. Rise onset was the first point at $Q \geq 50$ mAh cm^{-2} exceeding the 20 mAh cm^{-2} to 50 mAh cm^{-2} baseline median by four median absolute deviations for five consecutive grid points. The derivative maximum was searched only from 50 mAh cm^{-2} to $Q_{\text{end}} - 2.5$ mAh cm^{-2}. These deterministic ranges are estimator sensitivity, not confidence intervals. The derivative landmarks were not used to set Q_s or $Q_{\text{f},cal}$; $Q_{\text{f},cal}$ belongs to the pre-existing voltage-calibrated continuum branch. The same full-cell trace is used separately in the reduced observation-layer fit in Fig. S6, so this overlay is not independent validation.

Across the three windows, 11 of 12 rise onsets preceded the canonical modeled average-saturation marker, whereas derivative maxima spanned 92.0 mAh cm^{-2} to 115.25 mAh cm^{-2} and lay on both sides of the canonical calibrated half-accessibility coordinate. The experimental feature is therefore distributed rather than a single positive-electrode state validation.

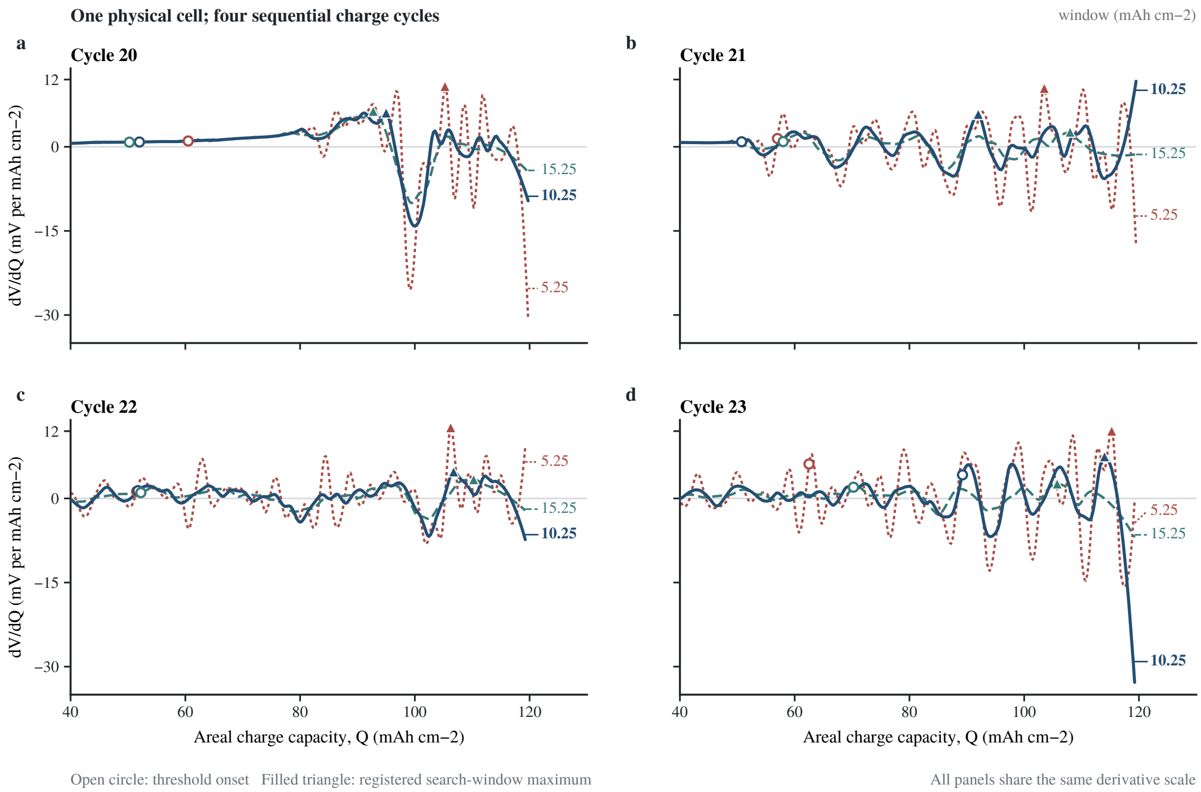

Figure S1: Derivative-window sensitivity within one physical cell. a–d, Full-cell charge- curve derivatives for eligible cycles 20–23 of one pristine cell. Direct labels identify the three prespecified physical windows; open circles mark registered rise onsets and filled triangles the registered maxima. Boundary fluctuations outside the maximum-search interval remain visible. All panels share one derivative scale. The cycles are within-cell repeated observations, not independent replicates. These full-cell features do not identify a positive-electrode internal state. Derived electrolyte labels follow `EXP-META-001`; raw records are unchanged.

Table S2: Voltage-rise onset and derivative maximum for every eligible cycle and smoothing window. The table reports deterministic estimator outputs for all cycle–window combinations rather than a selected landmark. Capacities are in mAh cm^{-2}.

cycle	window	rise onset	derivative maximum
20	5.25	60.50	105.25
20	10.25 (primary)	52.00	95.00
20	15.25	50.25	92.75
21	5.25	57.00	103.50
21	10.25 (primary)	50.75	92.00
21	15.25	58.00	108.00
22	5.25	51.50	106.25
22	10.25 (primary)	51.75	106.75
22	15.25	52.25	110.25
23	5.25	62.50	115.25
23	10.25 (primary)	89.25	114.00
23	15.25	70.25	105.75

S1.3 Descriptive supporting-electrolyte composition records

The 0–4 M supporting-electrolyte series was acquired under an approximately 80 mA total-current program (19 mA cm^{-2} to 21 mA cm^{-2} over 4 cm^2). Six physical cells were conservatively identified from 16 candidate paths and eight included run files. Continuations were stitched by acquisition identity and time gap; duplicate snapshots were not counted as new cells. Cycles were first summarized within each cell. No concentration law, optimum or population interval is fitted.

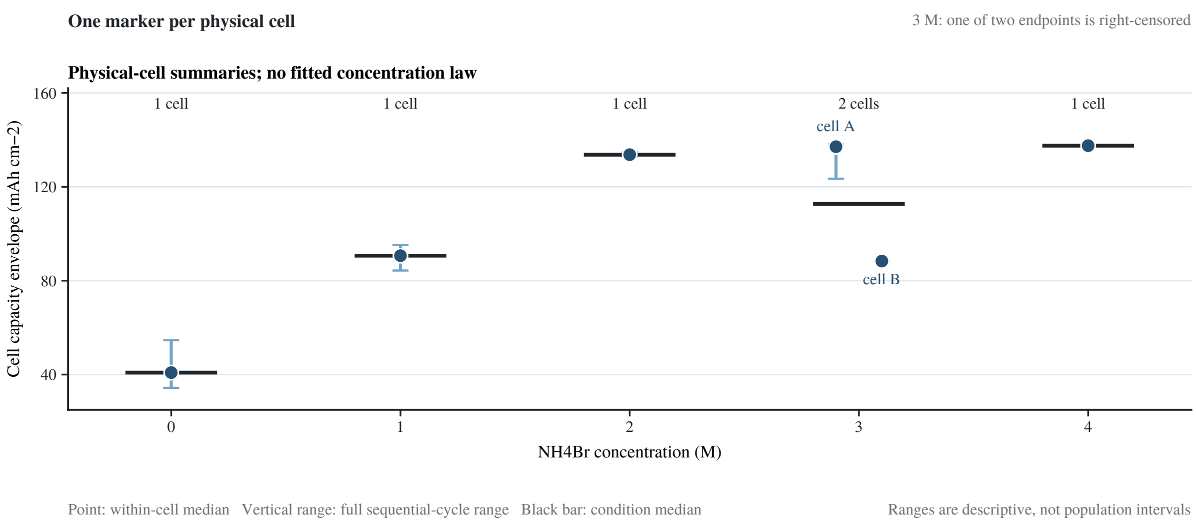

Figure S2: Cell-level descriptive capacity envelope across the existing composition records. Each filled marker is one physical cell and gives the maximum, across commanded levels, of the within-level median delivered capacity. Vertical segments are full sequential-cycle ranges at the selected command, and short black bars are condition medians. Physical-cell counts are 1/1/1/2/1 at 0/1/2/3/4 M NH4Br. The two acquisitions at 2 M form one conservatively stitched cell; the two 3 M records are distinct, and one 3 M endpoint is right-censored. The display is descriptive and does not identify an internal electrode state.

Table S3: Cell-level composition summaries and CE-qualified command brackets. The endpoint is the maximum of the within-command median delivered capacity under the primary 85% within-cell median-CE rule. Observed last-pass/first-fail levels are command-resolution brackets, not confidence intervals.

NH4Br (M)	n_{cell}	capacity envelope (mAh cm^{-2})	CE-qualified endpoint
0	1	40.841	[40, 60)
1	1	90.667	[80, 100)
2	1	133.696	[140, 160)
3	2	137.096; 88.354	[140, 160); ≥ 100 (right-censored)
4	1	137.539	[140, 160)

S1.4 Descriptive compression records

The 1.5, 2.5 and 3.0 mm records are explicit-thickness cells from the March 2025 campaign. The nominal 2.0 mm point is an implicit-standard-thickness cell from a different acquisition batch and remains an open, unconnected proxy. Each thickness represents one physical cell. The voltage-gap metric used in the source reconstruction is the energy-weighted mean charge–discharge gap, not a mid-capacity difference.

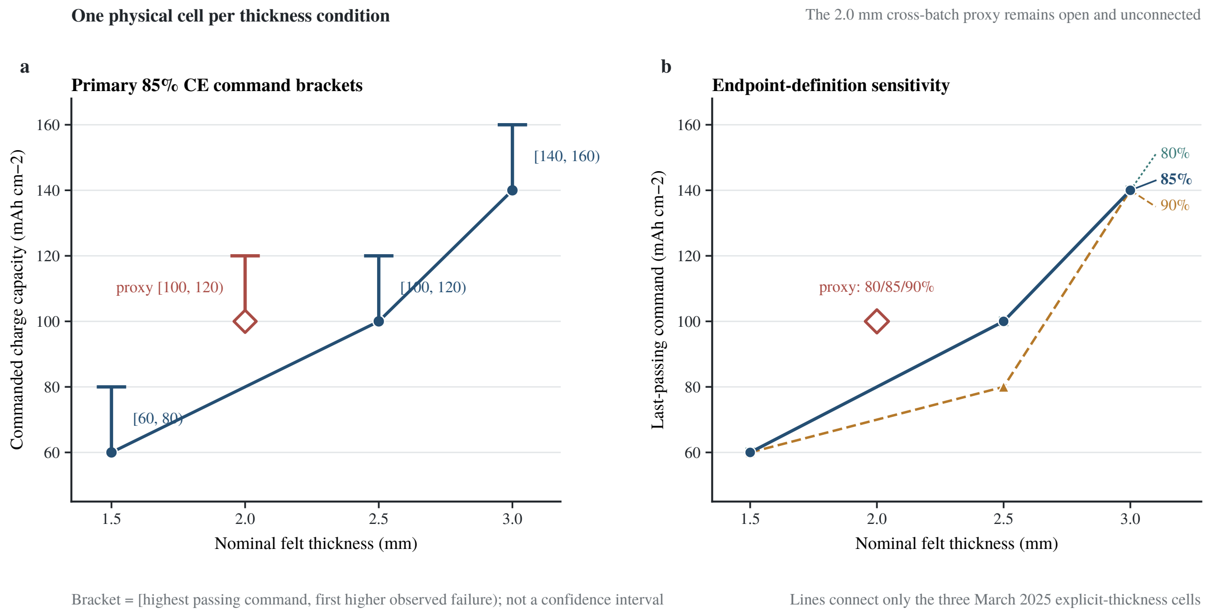

Figure S3: Protocol-bounded capacity brackets across existing felt-thickness cells. a, Primary 85% within-cell median-CE endpoint. The filled blue points connect only the three explicit March cells; the open 2.0 mm cross-batch proxy is unconnected. **b,** Prespecified 80%, 85% and 90% endpoint sensitivity. Each bracket is program resolution, not a confidence interval. The data do not define a population scaling law, pore-volume law or morphology relation.

Table S4: Compression-series inclusion, endpoint brackets and delivered capacity. Values use the primary 85% within-cell median-CE endpoint. Delivered capacity and CE are medians at the last passing level.

L (mm)	role	n_{cell}	last pass	first fail	delivered (mAh cm^{-2})	CE (%)
1.5	explicit	1	60	80	57.262	95.437
2.0	cross-batch proxy	1	100	120	95.271	95.271
2.5	explicit	1	100	120	85.956	85.956
3.0	explicit	1	140	160	135.511	96.793

S1.5 Rate and high-loading records

Only the 80 mA cm^{-2} to 360 mA cm^{-2} records form one same-cell pristine sequence. The 40 mA cm^{-2} pristine source and 400 mA cm^{-2} P350C-positive proxy are independent cells and remain unconnected. The pristine rate sequence is right-censored at 360 mA cm^{-2} with respect to a 95% CE-failure current.

These full-cell observations combine both electrodes, separator, shuttle and recovery losses. They can anchor and stress-test a positive-electrode model but cannot validate free iodine, retained-solid inventory or accessibility separately.

Table S5: Source roles and efficiency summaries for the rate records. Medians summarize sequential cycles within one physical cell at each current. The 40 and 400 mA cm^{-2} sources do not extend the pristine rate line.

source role	material identity	current (mA cm^{-2})	n_{cell}	n_{cycle}	median CE / EE (%)
cross-protocol marker	P-pristine/N-pristine	40	1	4	94.911 / 86.754
rate-ladder start	P-pristine/N-pristine	80	1	1	98.272 / 82.696
rate ladder	P-pristine/N-pristine	120	1	4	98.368 / 76.317
rate ladder	P-pristine/N-pristine	160	1	4	98.742 / 70.678
rate ladder	P-pristine/N-pristine	200	1	4	98.776 / 65.533
rate ladder	P-pristine/N-pristine	240	1	4	98.725 / 60.360
rate ladder	P-pristine/N-pristine	280	1	4	97.770 / 55.217
rate ladder	P-pristine/N-pristine	320	1	4	97.191 / 50.737
rate-ladder end	P-pristine/N-pristine	360	1	4	96.703 / 47.163
material proxy only	P350C/N-pristine	400	1	4	87.185 / 36.956

S2 Positive-electrode continuum formulation and computational identity

S2.1 State variables and reaction balances

The continuum calculation transports one lumped oxidized-iodine pool, c_O, rather than separate mobile $I_3{}^-$, I_2Br^- and aqueous I_2 fields. Remaining iodide is an algebraic stoichiometric inventory,

$$c_R = \max(c_{R,\text{floor}}, c_{R,0} - 3c_O), \tag{S1}$$

$$\varepsilon\, \partial_t c_O + \nabla\cdot(\mathbf{u}c_O - D_{O,\text{eff}}\nabla c_O) = R_\text{b} - R_\text{dep} + (r_\text{d} - r_\text{p})/V_m. \tag{S2}$$

The local speciation buffer is

$$\beta = 1 + K_\text{I}c_R + K_\text{Br}c_\text{Br}, \qquad c_{\text{I}_2}^{\text{free}} = c_O/\beta, \qquad S = c_{\text{I}_2}^{\text{free}}/c_{\text{sat}}^{\text{eff}}, \tag{S3}$$

with the implemented surface-effective stress also including the registered boundary-layer generation term. Thus c_R, $c_{\text{I}_2}^{\text{free}}$ and S are algebraic consequences of solved states and imposed closures, not additional transported fields.

Charge is conserved through

$$i_v = a_\text{b}j_\text{b} + a_s j_\text{dep}, \qquad \nabla\cdot\mathbf{i}_s = -i_v, \quad \nabla\cdot\mathbf{i}_l = i_v, \quad \mathbf{i}_s = -\sigma_{s,\text{eff}}\nabla\phi_s, \quad \mathbf{i}_l = -\kappa_{l,\text{eff}}\nabla\phi_l. \tag{S4}$$

The soluble path uses constant-exchange-current Butler–Volmer kinetics,

$$j_\text{b} = i_0\left[\exp\left(\frac{\alpha_a F\eta}{RT}\right) - \exp\left(-\frac{\alpha_c F\eta}{RT}\right)\right], \qquad \eta = \phi_s - \phi_l - E_\text{eq}, \tag{S5}$$

where the implemented E_eq is an empirical total-pool Nernst-like relation, not a complete activity model. A supersaturation-gated native deposited-solid path advances the retained-solid state,

$$j_\text{dep} = i_{0,\text{dep}}^{\text{base}}\max(S_\text{local} - 1, 0)10^{\eta/A_a}, \tag{S6}$$

$$\partial_t\varepsilon_\text{s} = V_m\frac{a_s j_\text{dep}}{2F}. \tag{S7}$$

The distinct auxiliary states satisfy

$$\partial_t\varepsilon_{\text{s,aux}} = r_\text{p} - r_\text{d}, \tag{S8}$$

$$\partial_t N_{\text{I}_2} = \kappa_N(N_\text{target} - N_{\text{I}_2}), \tag{S9}$$

and are not substituted for ε_s. The dimensionless N_{I_2} enters the calibrated relation

$$\theta_\mathrm{cal} = 1 - \exp\left[-6N_{\mathrm{I}_2}^{1/3}\varepsilon_{\mathrm{s,reg}}^{2/3}\right], \qquad R_\theta \equiv 1 - \theta_\mathrm{cal}. \tag{S10}$$

The native reaction-area identity contains a separate smooth pore factor,

$$T_\mathrm{pore} \equiv \left(\frac{K}{K_0}\right)^{1/2}, \qquad \frac{A_\mathrm{bare}}{A_0} = R_\theta T_\mathrm{pore}, \qquad a_\mathrm{b} = a_s R_\theta T_\mathrm{pore}. \tag{S11}$$

Consequently, $1 - \theta_\mathrm{cal}$ is R_θ, not the native A_bare/A_0 field. The calibrated half-accessibility coordinate $Q_{\mathrm{f},cal}$ is defined by $\langle\theta_\mathrm{cal}\rangle = 0.5$ (equivalently $\langle R_\theta\rangle = 0.5$), not by halving the product in Eq. (S11).

The remaining smooth feedback relations are

$$\varepsilon_{l,\mathrm{eff}} = \varepsilon - \varepsilon_\mathrm{s}, \qquad \frac{K}{K_0} = \left(\frac{\varepsilon_{l,\mathrm{eff}}}{\varepsilon}\right)^3 \left(\frac{1-\varepsilon}{1-\varepsilon_{l,\mathrm{eff}}}\right)^2, \qquad \{D, \kappa_l\} \propto \left(\frac{\varepsilon_{l,\mathrm{eff}}}{\varepsilon}\right)^{3/2}. \tag{S12}$$

Table S6: Model objects, equations, exported sources and evidence roles. Primary solved states, algebraic consequences, calibrated relations and prescribed inputs are distinguished explicitly.

object	source or equation	status	class	interpretation boundary
c_O	Eq. (S2)	transported solved state	E-SIM	lumped oxidized pool; species-resolved transport is not solved
c_R	Eq. (S2); `cI_m_surf_avg`	algebraic consequence	E-SIM	reconstructed iodide inventory, not a second solved species
$c_{\mathrm{I}_2}^\mathrm{free}$, S	Eq. (S3); `S_surf`	algebraic consequence	E-SIM	modeled free-iodine stress; not a measured concentration field
ε_s	Eq. (S7); `eps_s_pos`	native solved state	E-SIM	retained-solid fraction; not a deposit image
$\varepsilon_{\mathrm{s,aux}}$, N_{I_2}	Eq. (S9)	auxiliary ODE states	E-SIM	tracker and dimensionless dynamic state; neither replaces ε_s
θ_cal, R_θ	Eq. (S10); `theta_eff`	calibrated relation	E-SIM	calibration-controlled accessibility factor; not morphology or full native area
T_pore, A_bare/A_0	Eq. (S11); `A_bare_frac`	native algebraic closure	E-SIM	native area is $R_\theta T_\mathrm{pore}$; R_θ alone must not be relabelled as this field
porosity, K/K_0, D, κ_l	Eq. (S12)	smooth closures	E-SIM	mean-field transport channel; no local blockage is resolved
$\mathbf{u}$, p	`u_native_mag`, `p_native`	solved hydraulic fields	E-SIM	p is hydraulic pressure; no electrical-potential map is exported
j_total	Eq. (S4); `j_total`	solved reaction field	E-SIM	signed current is retained; non-positive nodes are not absolutized
terminal voltage	`voltage_V`	galvanostatic output	E-SIM	conditional on the calibrated model and full-cell boundary representation

S2.2 Geometry, flow, reservoir, boundary and initial conditions

The two-dimensional positive-electrode domain spans the compressed through-plane coordinate $x \in [3, 5]$ mm from current collector to separator and the flow-wise coordinate $y \in [0, 20]$ mm. Mechanical compression keeps felt-fibre inventory fixed,

$$(1-\varepsilon)L = 0.30\,\mathrm{mm}. \tag{S13}$$

Electrolyte flow obeys Darcy's relation

$$\mathbf{u} = -\frac{\kappa_0(K/K_0)}{\mu}\nabla p, \qquad \nabla\cdot\mathbf{u} = 0, \tag{S14}$$

and is coupled to the electrochemical calculation through convection. The registered outlet reference is $p_{\mathrm{out}} = 0$; all displayed pressure is hydraulic pressure in pascals. Only the solved loading window is interpreted. No mechanical-failure or deep extrapolation is assigned physical meaning.

A well-mixed reservoir of volume V_{tank} supplies the transported pool,

$$V_{\mathrm{tank}} \frac{\mathrm{d}c_O^{\mathrm{tank}}}{\mathrm{d}t} = \dot{V}(c_O^{\mathrm{out}} - c_O^{\mathrm{tank}}). \tag{S15}$$

At the inlet, $c_O = c_O^{\mathrm{tank}}$ and the superficial flow is prescribed. The outlet has zero diffusive species flux and $p = p_{\mathrm{out}}$. Iodine species have zero normal flux at collector and separator; electronic current enters at the collector and ionic current leaves at the separator. Initial c_O is uniform, c_R follows Eq. (S2), and $\varepsilon_{\mathrm{s}} = 0$.

S2.3 Parameter roles

Table S7: Cell, geometry, protocol and electrochemical inputs. Values are classified as recorded hardware/protocol, prescribed scenario, calibrated relation or numerical model choice.

quantity	symbol	value	role and provenance
electrode area	A	$4\,\mathrm{cm}^2$	recorded hardware
felt thickness; porosity rule	$L, \varepsilon(L)$	$2\,\mathrm{mm}$; $1 - 0.30\,\mathrm{mm}/L$	prescribed baseline and structural closure
specific area	a_s	$4 \times 10^4\ \mathrm{m}^{-1}$	empirical geometry scale; no independent area measurement
separator	—	Nafion 212 ($\approx 50\,\mu\mathrm{m}$)	recorded hardware
tank volume; flow	$V_{\mathrm{tank}}, \dot{V}$	$25\,\mathrm{mL}$; $50\,\mathrm{mL\,min}^{-1}$	prescribed baseline protocol
temperature	T	$293.15\,\mathrm{K}$	continuum-model setting; not assigned to experimental files
applied current density	i_{app}	$400\,\mathrm{A\,m}^{-2}$ ($40\,\mathrm{mA\,cm}^{-2}$)	prescribed baseline
electrolyte condition	—	1 M ZnI_2 + 4 M NH4Br	representative supporting-electrolyte condition
formal potential	$E^{0\prime}$	$0.548\,\mathrm{V}$	prescribed reference; no present-electrolyte anchor
exchange current	i_0	$5 \times 10^4\ \mathrm{A\,m}^{-2}$	fast-kinetics closure; not an intrinsic measurement
transfer coefficients	α_a, α_c	1, 1	prescribed symmetric Butler–Volmer relation
liquid conductivity	κ_l	$20\,\mathrm{S\,m}^{-1}$	undocumented prescribed value; no raw conductivity record located
Nernst-like reference	c_{ref}	$120\,\mathrm{mol\,m}^{-3}$	empirical total-pool reference

At the canonical endpoint, the native retained-solid fraction is 3.21926×10^{-3} while the auxiliary tracker is 1.08328×10^{-6}. The native path carries 0.5516% of integrated charge and 27.994% of the endpoint positive current. These values demonstrate that the two states are not interchangeable.

Table S8: Transport, speciation and saturation inputs. Transport and speciation values are lower-scale or literature priors unless a present-electrolyte measurement record is identified.

quantity	symbol	value	role and provenance
free-I_2 diffusivity	D_{I_2}	0.6×10^{-9} m^2 s^{-1}	lower-scale mobility prior; porous effective value remains model dependent
polyiodide diffusivity	$D_{I_3^-}, D_{I_2Br^-}$	0.5×10^{-9} m^2 s^{-1}	force-field-limited lower-scale prior
mass-transfer length	δ	25 µm	prescribed sub-grid scenario; not measured
deposit-free permeability	κ_0	1.0×10^{-10} m^2	prescribed felt-scale value
viscosity; density	μ, ρ_l	2.2×10^{-3} Pa s; 1300 kg m^{-3}	prescribed hydraulic values; not measured here
iodide complexation	K_I	0.72 m^3 mol^{-1}	selected conditional prior near the 698 ± 10 M^{-1} scale reported by Palmer *et al.* [2]
bromide complexation	K_{Br}	0.0115 m^3 mol^{-1}	rounded conditional prior near 12.2 L mol^{-1} from Lee *et al.* [3]; system-specific formation precedent is not used as this number
intrinsic aqueous-I_2 solubility	c_{sat}	1.33 mol m^{-3}	dilute-water reference [4]
salting factor	γ_{salt}	3; $c_{sat}^{eff} \approx 0.44$ mol m^{-3}	bounded cross-salt prior, varied from 2 to 4 [2]
I_2 molar volume	V_m	5.15×10^{-5} m^3 mol^{-1}	conversion from registered molar mass and density

Table S9: Native-solid pathway, auxiliary-state constants and numerical regularizers. The native solid inventory and auxiliary dynamic states have distinct equations and evidential roles.

quantity	symbol	value	role
native-solid exchange scale	$i_{0,dep}^{base}$	0.05 A m^{-2}	empirical deposited-species path
native-solid Tafel slope	A_a	118 mV	prescribed path parameter
auxiliary formation/dissolution	k_p, k_d	$10^{-9}, 5{\times}10^{-9}$ m s^{-1}	empirical auxiliary-rate closures
dynamic-state relaxation	κ_N	10^{-2} s^{-1}	calibrated relation
dynamic-state scales	$N_{ref}, N_{sup}, N_{max}$	$1, 500, 2 \times 10^7$	dimensionless calibrated-state constants
overpotential scale	η_{nuc}	3 mV	calibrated dynamic-state parameter
supersaturation smoothing	—	0.02 mol m^{-3}	numerical smoothing
seed floor	ν_0	0.03	phenomenological initiation floor
solid-fraction regularizers	$\varepsilon_s^{fl}, \varepsilon_s^{sm}$	$10^{-9}, 10^{-8}$	numerical regularizers
initial felt porosity	ε_0	0.85	prescribed structural scenario

S2.4 Computational identity and numerical provenance

COMSOL Multiphysics 6.3.0.290 (COMSOL AB, Stockholm, Sweden) was used for the continuum calculations. The baseline figures retain the registered R526 production identity, while the matched accessibility comparison uses the convergence-qualified true-mesh pair.

Table S10: Continuum-model identity, input hashes and numerical checks. Original and archived solved models were not modified. Matched calculations began from byte-identical copies.

object	software object	immutable identity or setting	check
canonical baseline	branch `PB-R526-J40-Q120-DSET5-v1`	study `stdR522`; solution `sol5`; dataset `dset5`; 1081 exported points over `range(0,10,10800)`	source of baseline fields and canonical event coordinates
source input copy	two independent copied `.mph` files	`4813B306F39330CCEAF819C4AB8815C5` `5A3DF8A04726741F33ACB801B8806B9B`	byte-identical before model-form change
fresh control	`stdR522/sol5/dsetR581Ctrl`	solved copy SHA-256 `B26D8FA00159C7EF` `90B5932886C54F285BBBD8EBA899A0F7` `C62AAD7D8FBC6FED`	maximum voltage difference from historical trace: 0.285 mV
physical-dense copy	`stdR522/sol5/dsetR581Phys`	only `cov_theta_surf` and `theta_eff_R520` changed to $1 - \exp(-35.4\varepsilon_s^{0.6222})$	all parameter inventories byte-identical
tolerance pair	1944 elements; rtol = 3×10^{-4}	separate control and physical copied solves	predeclared tolerance gate passed
true-mesh pair	7776 elements; rtol = 3×10^{-4}; minimum quality 0.198	control timeseries SHA-256 `B6C0EAB603D17E94` `77CADAB04F6F0093D0C98261925561E5` `10C82B9203704569`; physical timeseries `22B8AE22303A850BAEA710E293992103` `C5783AB158B63F7E7ACD757561014AF8`	fourfold element-count gate passed
mesh response gate	true mesh versus same-tolerance 1944-element pair	ΔQ_s = 0.0126 (control), 0.1431 mAh cm^{-2} (physical); endpoint ΔV change 0.0938 mV	all predeclared state and identity checks passed

S3 Extended positive-electrode state fields

S3.1 Snapshot and event definitions

Exact exports at Q = 0, 80, 96, 100, 110 and 120 mAh cm^{-2} expose the progression before, between and after the canonical averaged event coordinates. The Q = 100 export is close to, but not identical with, $Q_{f,cal}$ = 99.5901 mAh cm^{-2}. Local $S > 1$ and native solid can occur before the electrode- average Q_s crossing.

Table S11: Event definitions and exported snapshot roles. Display capacities are exact exports; Q_s and $Q_{f,cal}$ are interpolated electrode-average coordinates.

coordinate (mAh cm^{-2})	registered definition	display role
Q_s = 83.0202	first upward crossing of $\langle S \rangle = 1$	canonical averaged saturation marker; not first local supersaturation
$Q_{f,cal}$ = 99.5901	$\langle \theta_{cal} \rangle = 0.5$, equivalently $\langle R_\theta \rangle = 0.5$	calibrated accessibility coordinate; not $A_{bare}/A_0 = 0.5$
0	exact export	start of charge
80	exact export	immediately before the average-saturation coordinate
96	exact export	between Q_s and $Q_{f,cal}$
100	exact export	nearest registered snapshot to $Q_{f,cal}$
110	exact export	post-half-accessibility progression
120	exact export	canonical analysis endpoint

S3.2 State, function and reaction fields

Figure S4 assembles the six registered positive-electrode exports using fixed row-wise scales and nearest-neighbour rendering without image smoothing.

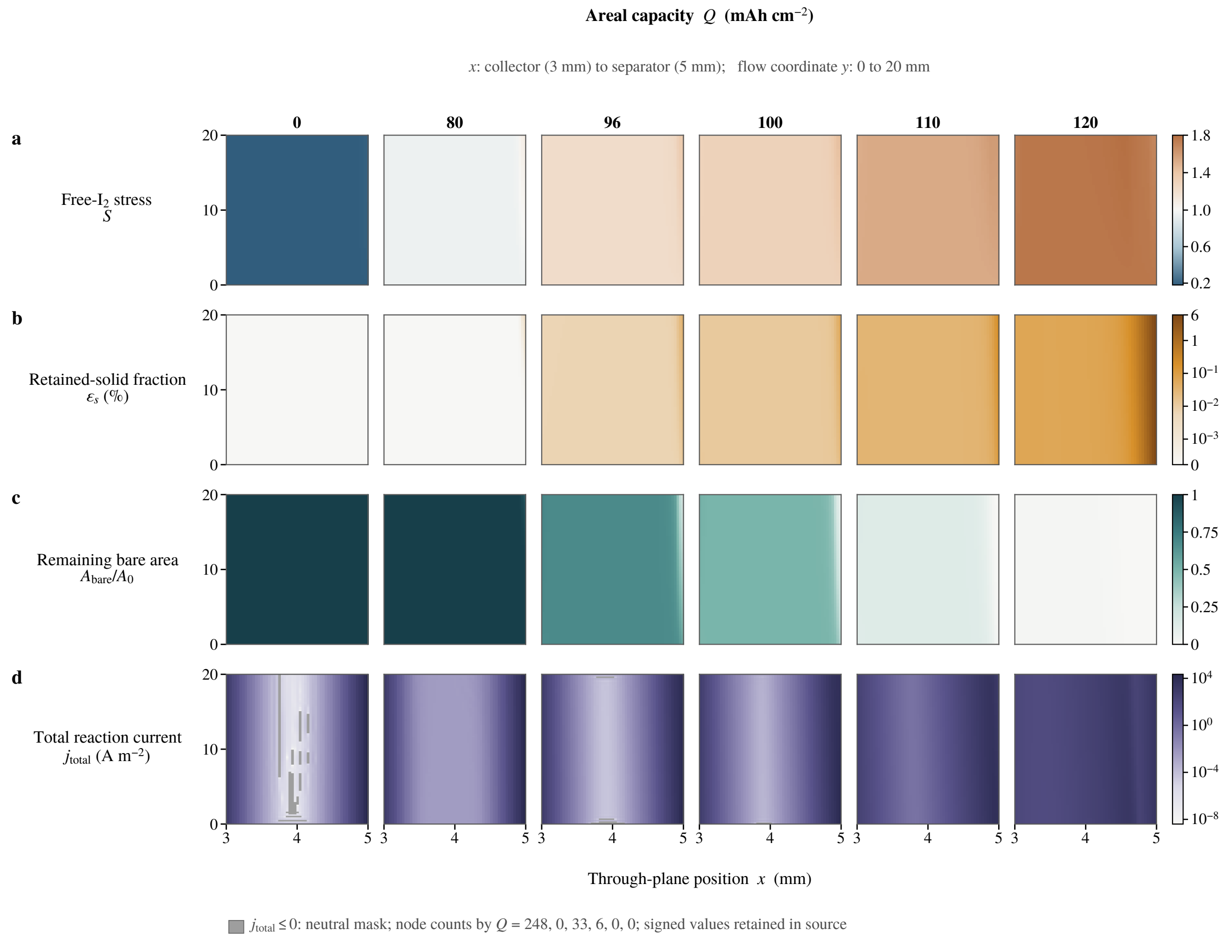

Figure S4: Six-capacity inventory of modeled positive-electrode state and reaction fields. Columns show $Q = 0$, 80, 96, 100, 110 and 120 mAh cm^{-2}. **a,** Free-I$_2$ stress S on a shared scale centred at $S = 1$. **b,** Native retained-solid fraction ε_s on a shared symlogarithmic percentage scale. **c,** Native remaining reaction-area fraction $A_{\mathrm{bare}}/A_0 = R_\theta T_{\mathrm{pore}}$ from the registered `A_bare_frac` export. **d,** Total reaction-current density. Positive values use one logarithmic scale; non-positive nodes are neutrally masked without taking absolute values. Their counts are 248, 0, 33, 6, 0 and 0 in column order, and signed values remain in the source table. Each map contains 5995 nodes and uses nearest-neighbour rendering without smoothing or clipping. These fields are model outputs, not microscopy, deposit morphology, a blockage front or electrical potential.

S3.3 Hydraulic fields

Figure S5 reports the corresponding smooth-permeability, velocity and hydraulic- pressure exports on fixed row-wise scales.

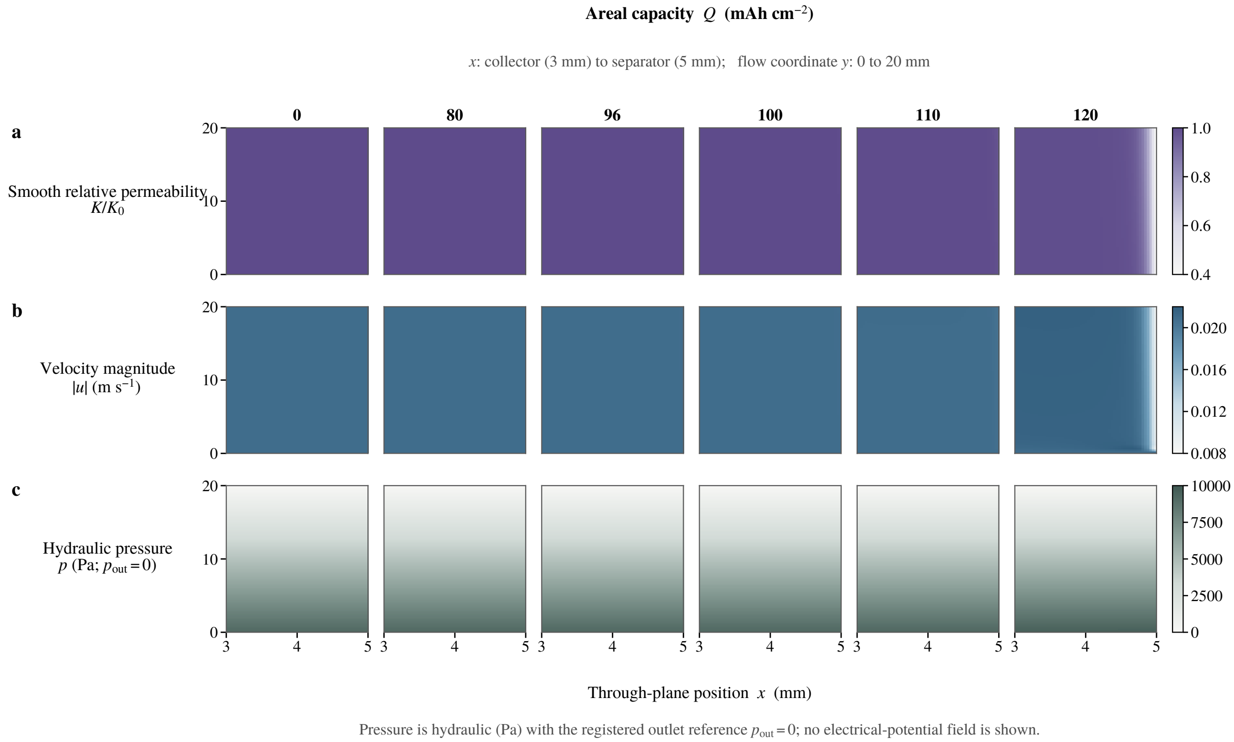

Figure S5: Hydraulic response across the same six positive-electrode states. a, Smooth relative permeability K/K_0. **b,** Native velocity magnitude in m s^{-1}. **c,** Hydraulic pressure in Pa relative to $p_{\text{out}} = 0$. Each row has one fixed scale across all capacities, and every map contains the same 5995 exported nodes. The pressure field is hydraulic, not an electrical potential. The smooth permeability channel does not resolve local pore closure or blockage.

S4 Accessibility-relation comparison, convergence and voltage identifiability

S4.1 Matched model-form sensitivity

The matched control and physical-dense calculations start from byte-identical copies. The control changes no expression; the coupled physical-dense calculation changes only the two registered accessibility expressions to the dense single-fibre comparator. A separate one-way shadow evaluates that comparator on the control $\varepsilon_s(Q)$ trajectory without feedback. These are three different objects.

Table S12: Convergence-qualified matched accessibility comparison. Changing only the accessibility relation leaves the average saturation point nearly unchanged but alters later accessibility, solid inventory and voltage. This is a controlled model-form sensitivity, not real-material causality or morphology identification. Endpoint values are at $120\,\mathrm{mAh\,cm^{-2}}$.

quantity	true-mesh control	one-way dense shadow	coupled physical dense
Q_s ($\mathrm{mAh\,cm^{-2}}$)	82.8903	inherited control state	82.8924
$Q(\theta_{\mathrm{cal}} = 0.5)$ ($\mathrm{mAh\,cm^{-2}}$)	99.4637	118.6927	not reached
endpoint V (V)	1.728364	not a solve	1.440208
endpoint ε_{s}	3.15236×10^{-3}	inherited control state	6.58584×10^{-4}
endpoint θ	0.989982	0.625898	0.309081
endpoint $R_\theta = 1 - \theta$	0.0100178	0.374102	0.690919
endpoint A_{bare}/A_0	0.00996125	not evaluated as a native solve	0.687120
endpoint K/K_0	0.957035	inherited control state	0.988976
$V_{\mathrm{physical}} - V_{\mathrm{control}}$		−288.156 mV	
mesh-gate change in this difference		0.0938 mV	

S4.2 Observation-layer identifiability

The reduced observation layer fits a selected full-cell voltage interval with a baseline, linear capacity term and one accessibility-related logarithmic contribution. It is an identifiability diagnostic, not a second continuum solve. In this reduced layer, the varied accessibility coordinate is R_θ; it is not the native $A_{\mathrm{bare}}/A_0 = R_\theta T_{\mathrm{pore}}$ field.

No internal-overpotential decomposition is used in the submission SI because it would add a conditional mechanism separation not required by the R582 claims.

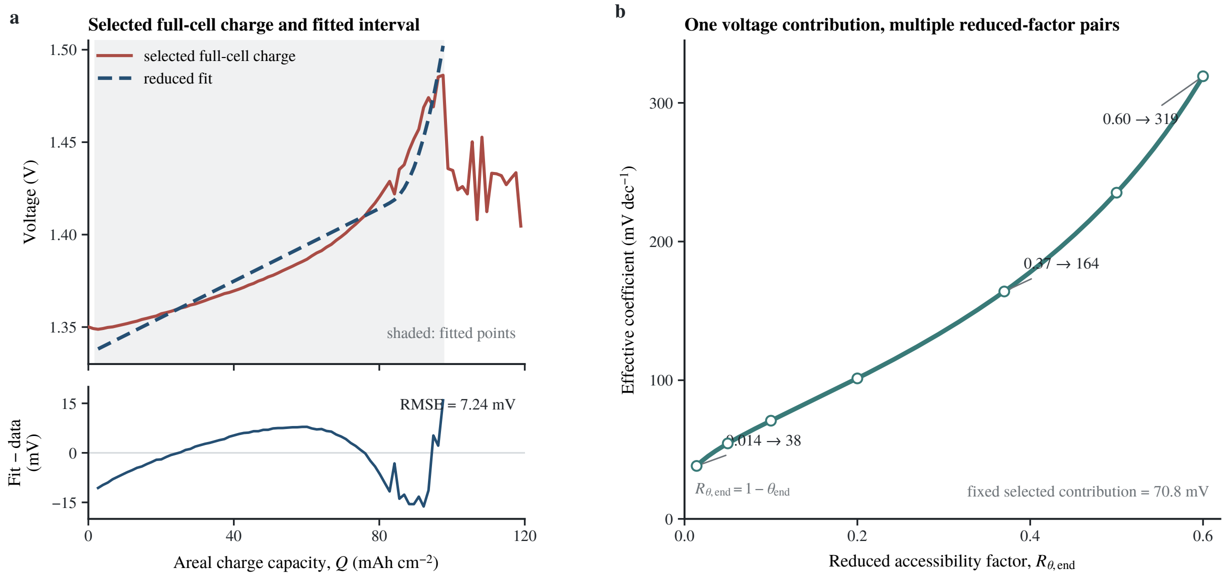

Figure S6: Full-cell voltage fit and accessibility–coefficient degeneracy. a, Selected pristine full-cell charge trace and reduced observation-layer fit. The shaded interval contains 72 points from 2.672 to 97.540 mAh cm^{-2} in $V = c_0 + c_1 Q + b_{\mathrm{eff}} \ln(1/R_\theta)$; the lower axis shows fit minus data. The fit gives $c_0 = 1.335\,54$ V, $c_1 = 9.819 \times 10^{-4}$ V (mAh cm^{-2})$^{-1}$, $b_{\mathrm{eff}} = 0.136\,22$ V and RMSE 7.24 mV. **b,** End-point R_θ–coefficient pairs that preserve the selected 70.813 mV accessibility-related voltage contribution. Open circles are registered pairs and the line is the analytical relation. Here R_θ is a reduced observation-layer factor, not native remaining reaction area. Voltage therefore does not identify the inventory-to-accessibility relation or deposit morphology uniquely.

S5 Molecular bounds on placement and mobility

S5.1 Periodic molecular-iodine calculations

Periodic electronic-structure calculations used CP2K/Quickstep 2026.1 [5, 6], PBE [7], D3(BJ) [8, 9], DZVP-MOLOPT-SR-GTH basis sets [10], GTH pseudopotentials [11–13], a 150 Ry density cutoff, a 30 Ry relative cutoff and Γ-point sampling. Counterpoise-corrected single-point energies define the single-I_2 ordering [14]. These finite-periodic-cell electronic energies are not solution free energies, site populations or nucleation barriers.

Table S13: Periodic single-I_2 BSSE-corrected adsorption energies and two-I_2 compact-versus-separated electronic-energy diagnostic. Single-I_2 values are BSSE-corrected adsorption energies within the declared calculations; two-I_2 relative values compare the registered compact and separated configurations. Both are used only as placement tendencies.

calculation	BSSE-corrected adsorption energy (eV)	relative reference (eV)	permitted use
single I_2, C–OH	−0.997724	−0.595122 versus basal	site-ordering prior
single I_2, C=O	−0.848237	−0.445635 versus basal	site-ordering prior
single I_2, basal	−0.402602	0	reference
single I_2, vacancy	−0.318468	+0.084134 versus basal	site-ordering prior
compact two I_2 at C–OH	—	−0.689291 versus separated	configuration diagnostic
separated two-I_2 reference	—	0	reference

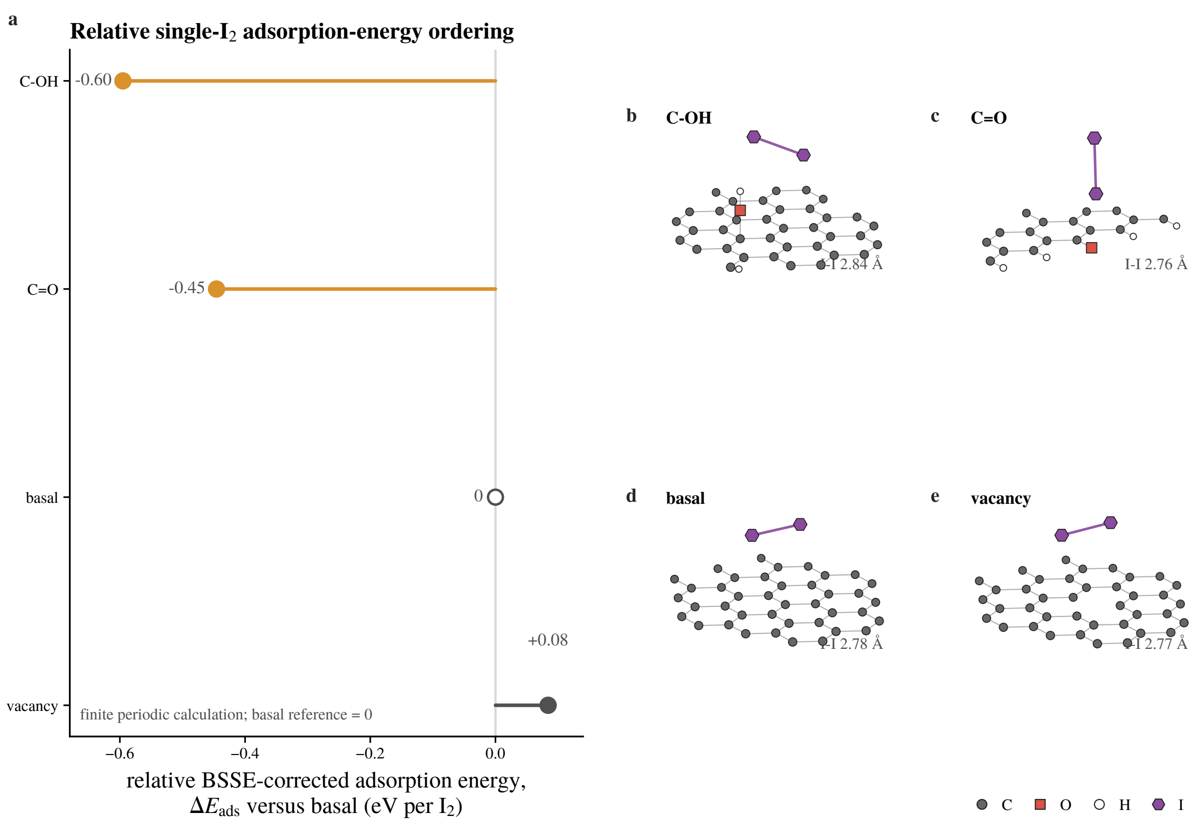

Figure S7: Relative single-I_2 BSSE-corrected adsorption-energy ordering and optimized geometries. a, BSSE-corrected adsorption energies relative to the basal calculation. **b–e,** Exact optimized geometries in the same site order; displayed adjacency guides are distance based and do not assign bond order. Within the declared finite periodic calculation, the tested C–OH and C=O motifs lie below basal carbon and the vacancy lies slightly above it. The unavailable raw single-I_2 charge-density fields are not used.

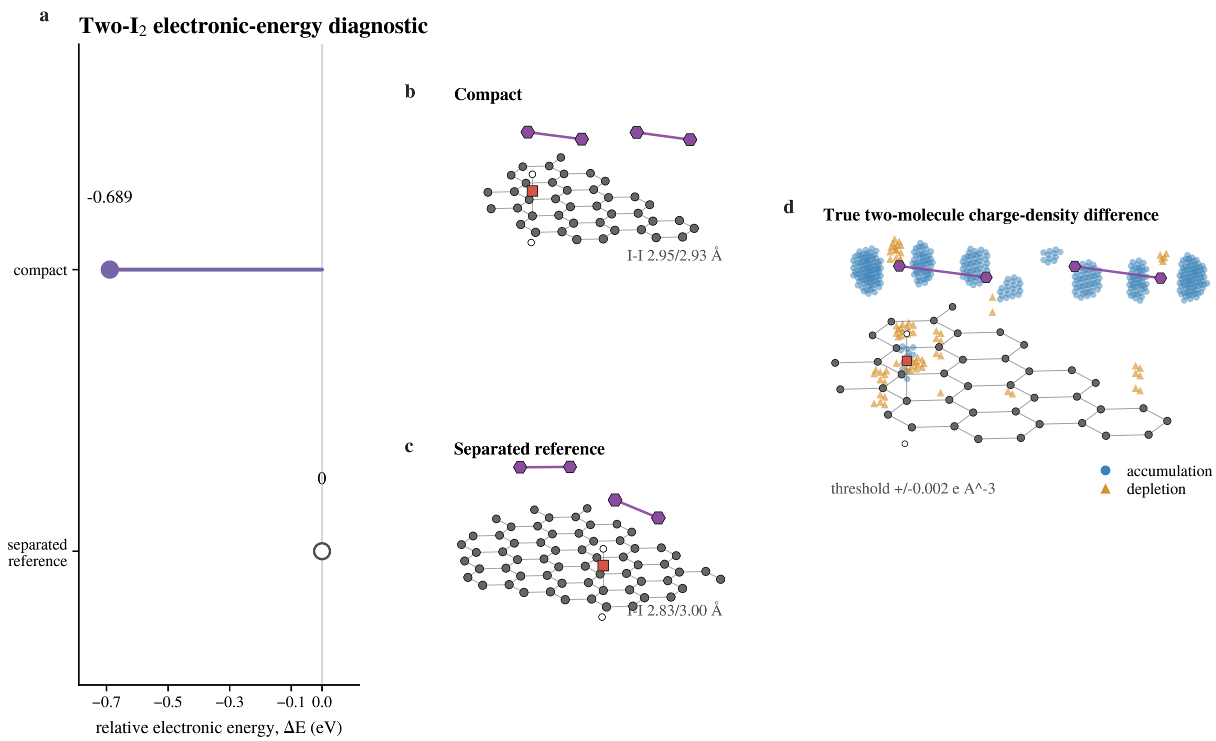

Figure S8: Compact and separated two-I_2 configurations at the tested C–OH site. a, Registered compact-minus-separated electronic-energy comparison. **b,c,** Exact optimized compact and separated geometries. **d,** Recoverable two-I_2 charge-density-difference field, thresholded at $\Delta\rho \geq +0.002$ or $\leq -0.002\ e\ \text{Å}^{-3}$ without smoothing or interpolation; 428 accumulation and 78 depletion points are retained. One periodic translation is used only to display the compact cluster continuously. The compact configuration is 0.689 291 eV below the separated reference within this calculation. The field is a diagnostic, not a reaction coordinate, solution equilibrium or nucleation pathway.

S5.2 Molecular-dynamics carrier prior

Classical MD used GROMACS 2026.0-conda_forge [15], SPC/E water [16], Joung–Cheatham-family halides [17], Packmol initial configurations [18], and project-specific analogue parameters for the oxidized carriers. Electronic-continuum charge scaling was compared with formal charges [19]. Five state-of- charge compositions were run once under each charge parameterization. Four contiguous trajectory blocks quantify within-trajectory stability and are not independent replicates.

Table S14: MD protocol, estimator and force-field boundaries. The protocol supplies a bounded carrier- mobility prior; block statistics quantify within-trajectory stability only.

item	frozen implementation	evidence boundary
software and cases	GROMACS 2026.0-conda_forge; five SOC compositions at q = 0.8 ECC and q = 1.0	one trajectory per composition and parameterization; ten cases are not replicates of one condition
trajectory	0.1 ns NVT, 0.5 ns NPT, 20 ns NVT production; 2 fs step; 5 ps frames	298.15 K; V-rescale thermostat [20], C-rescale barostat in NPT [21], PME electrostatics [22], 1.0 nm real-space cutoffs
diffusion estimator	molecular-centre-of-mass Einstein MSD; lag fit 2.51 ns to 8.49 ns	four contiguous 5 ns blocks provide a stability diagnostic, not replica SEM
base force fields	SPC/E water; Joung–Cheatham-style halides; SPC/E-compatible 12–6 Zn^{2+}; compact $NH_4{}^+$; neutral two-site I_2	several ion and molecular models are project approximations; no present-electrolyte force-field validation is claimed
polyhalides	GFN2-xTB/ALPB-water charge priors [23, 24]; manual bonded and analogue Lennard–Jones terms	absolute $I_3{}^-/I_2Br^-$ diffusivities are high-uncertainty
viscosity	18 planned Green–Kubo replica rows	required energy outputs are absent; these rows provide no result or uncertainty estimate
continuum use	bulk mobility ordering and range	porosity/tortuosity treatment is still required before use as D_{eff}

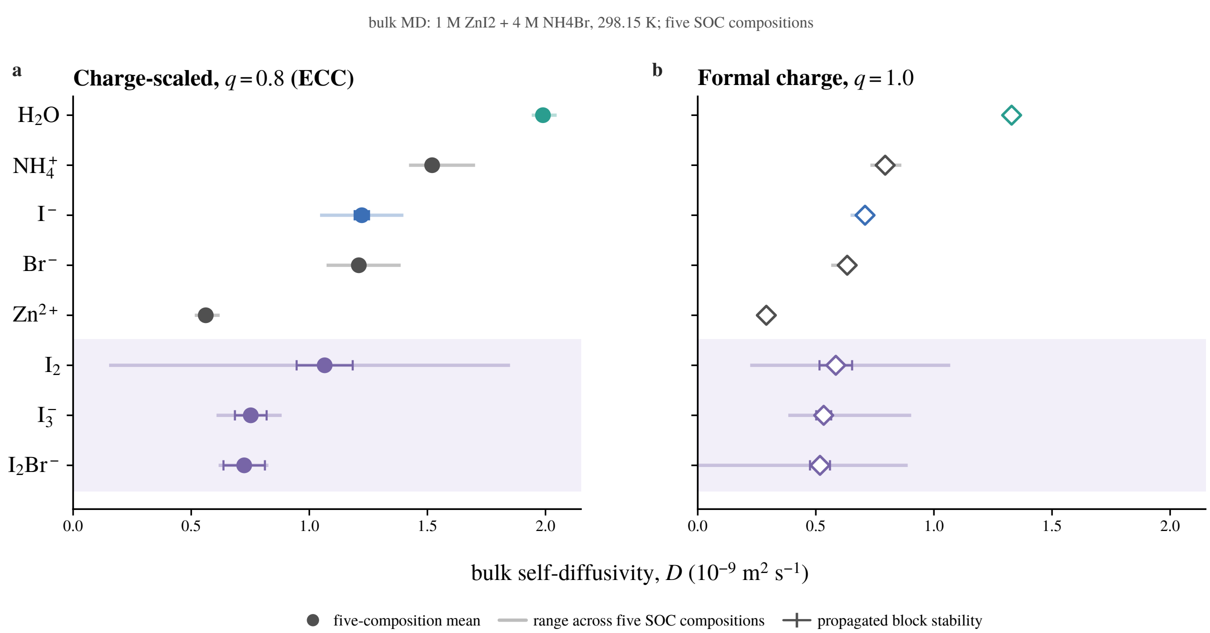

Figure S9: Full carrier-mobility ladder under charge-scaled and formal-charge parameterizations. a, q = 0.8 ECC. **b,** q = 1.0 formal charges. The condition is 1 M ZnI_2 + 4 M NH4Br at 298.15 K. Symbols are means across five different SOC compositions; thin spans show their min–max range and capped bars the propagated four-block within-trajectory stability measure. Neither is replica uncertainty. Oxidized-iodine carrier mobilities lie below iodide in both parameterizations, while absolute values remain force-field dependent.

S6 Single-fibre and pore-network comparator definitions

S6.1 Idealized domains and boundary conditions

Figure S10 records the prescribed single-fibre and pore-network domains used for the deterministic comparator calculations.

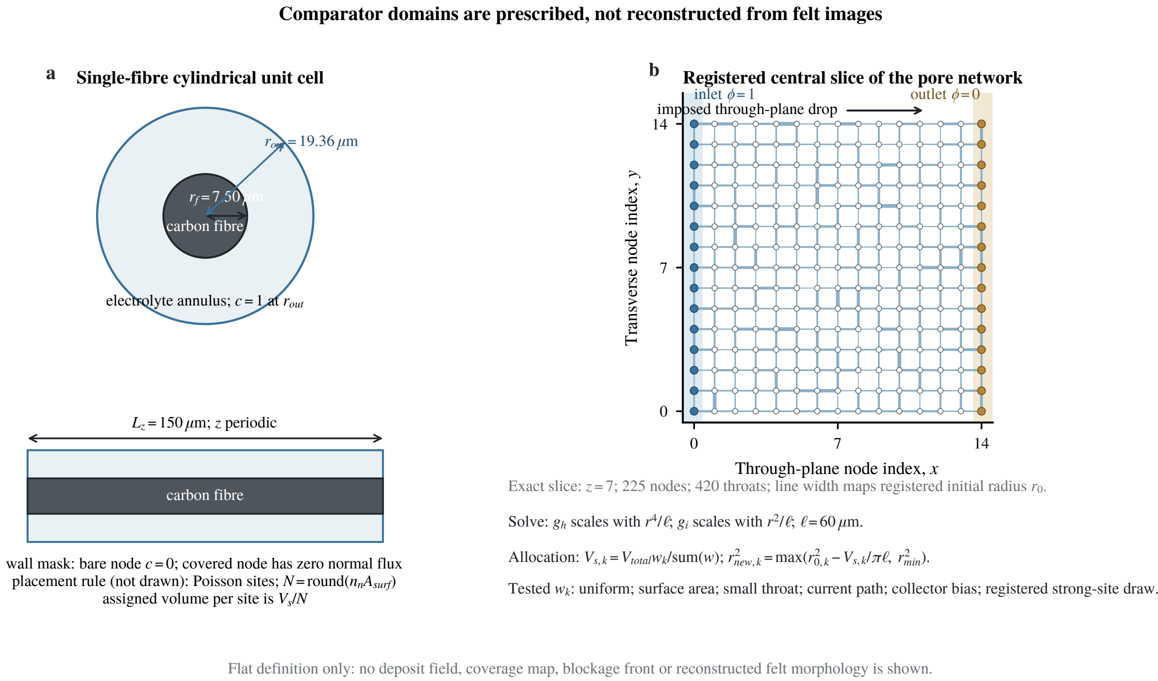

Figure S10: Flat definitions of the single-fibre and pore-network comparator models. a, Orthographic views of the prescribed cylindrical unit cell ($r_f = 7.50\,\mu$m, $r_{\text{out}} = 19.36\,\mu$m, $L_z = 150\,\mu$m, $\varepsilon = 0.85$). Normalized concentration is fixed at one at the outer radius; a bare wall node has $c = 0$, a covered node has zero normal flux, and azimuthal/axial directions are periodic. **b,** Exact $z = 7$ slice of the registered $15 \times 15 \times 15$ cubic network (225 nodes and 420 in-slice throats). Line width maps initial throat radius; opposing faces have normalized Dirichlet values one and zero. The 60 µm lattice and displayed equations define conductance, allocation and radius update. These are prescribed comparator domains, not images or reconstructions of the felt. No deposit field, solved pressure, electrical potential, flux or blockage front is shown.

S6.2 Accessibility families and threshold definitions

The isolated-fibre calculations vary a prescribed placement density and precipitation fraction. Their geometric A/A_0 is different from the continuum calibration coordinate R_θ. The latter can be plotted only as a separately labelled comparator; it is not a native A_{bare}/A_0 curve.

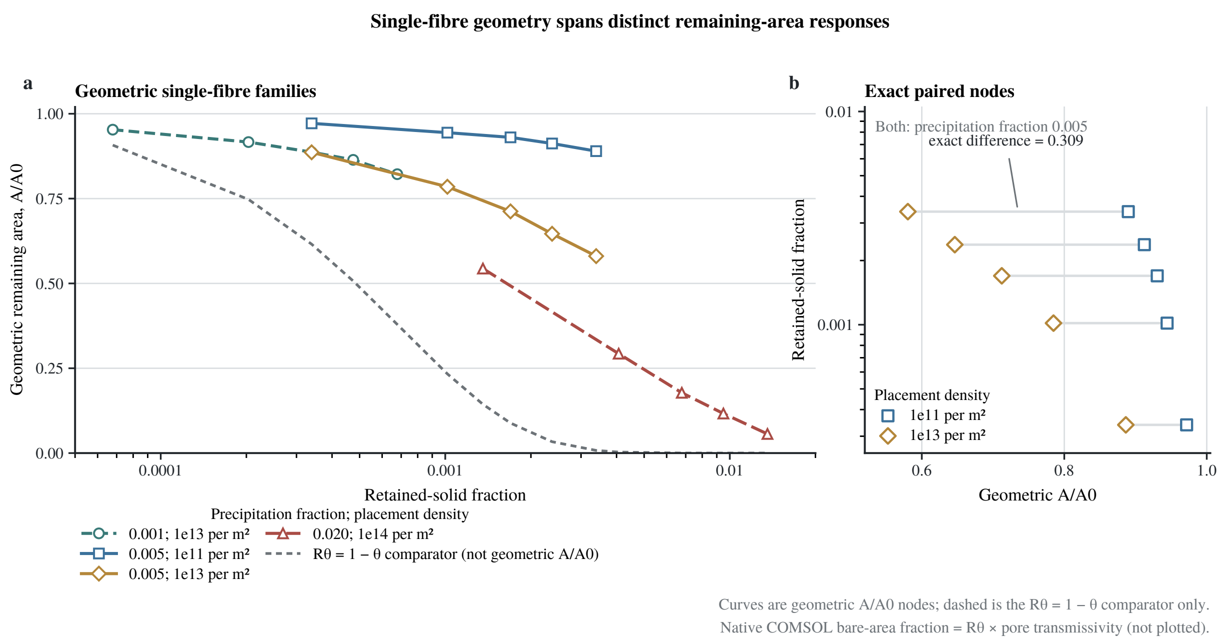

Figure S11: Remaining-accessibility families across the full tested placement-density range. a, Geometric single-fibre A/A_0 for all four registered comparator families. Lines connect five computed nodes without fitting. The dashed grey curve is the separately calibrated continuum R_θ relation sampled at the registered inventory nodes; it is not a geometric family and not native A_{bare}/A_0. **b,** Five shared inventory nodes for the two $\phi_{\mathrm{ppt}} = 0.005$ families. Increasing prescribed placement density from 10^{11} to 10^{13} m^{-2} changes geometric remaining area by 0.085–0.309 over $\varepsilon_{\mathrm{s}} = 3.39 \times 10^{-4}$ to 3.39×10^{-3}. These deterministic comparators are not measured coverage, felt morphology or validation of the calibrated relation.

Table S15: Definitions of calibrated, island-model and film-connectivity coordinates. These quantities have non-interchangeable definitions.

coordinate	value in ε_{s}	definition	boundary
calibrated trajectory readout	1.18572×10^{-4}	production trajectory when $\langle\theta_{\mathrm{cal}}\rangle = 0.5$	voltage-calibrated readout; not a material threshold and not $A_{\mathrm{bare}}/A_0 = 0.5$
dense-island half-area	1.7973×10^{-3}	$\theta_{\mathrm{island}} = 0.5$ for $1 - \exp(-35.4\varepsilon_{\mathrm{s}}^{0.6222})$	idealized single-fibre comparator; direct discrete interpolation is approximately 1.91×10^{-3}
film-connectivity marker	2.2324×10^{-3}	registered geometric connected-film/percolation criterion	not a half-accessibility definition

S7 Operating-variable scans and transport-channel controls

S7.1 Complete scan matrix

Each scan family changes one declared operating or model setting while holding its own remaining settings fixed. Absolute endpoint voltages are not ranked across families with different fixed operating points. Q_s is obtained by linear interpolation across the first adjacent samples that bracket $\langle S \rangle = 1$; $Q_{\mathrm{f},cal}$ is obtained analogously from $\langle \theta_{\mathrm{cal}} \rangle = 0.5$. Endpoint states in Table S16 report native ε_{s} and $R_\theta = 1 - \theta_{\mathrm{cal}}$; full native reaction area additionally contains T_{pore}.

Table S16: Tested operating/model-variable ranges and positive-electrode responses. Lists follow increasing setting. "NR" means the threshold was not reached within the stated solved trajectory. End states are reported at 120 mAh cm^{-2} unless another endpoint is stated. The analytical row is postprocessed, not a solved scan.

family	tested settings	Q_s (mAh cm^{-2})	$Q_{\mathrm{f},cal}$ (mAh cm^{-2})	endpoint ε_{s}; endpoint R_θ	class
applied current	20, 40 mA cm^{-2} to $Q = 120$; 80, 120 mA cm^{-2} to $Q = 40$	106.320; 83.020; > 40; 8.391	NR; 99.590; NR; 31.072	1.402×10^{-4}; 0.52485 (20, $Q = 120$). 3.219×10^{-3}; 0.01049 (40, $Q = 120$). 2.494×10^{-7}; 0.99881 (80, $Q = 40$). 4.793×10^{-4}; 0.22052 (120, $Q = 40$)	E-SIM
oxidized-carrier D_{eff}/D_0	0.5, 0.6, 0.8, 1.0, 1.2, 1.5, 2.0	45.034; 57.360; 73.131; 83.020; 89.972; 97.490; 106.205	61.099; 73.525; 89.503; 99.590; 106.748; 114.589; NR	ε_{s}: 0.166916; 0.108550; 0.032906; 0.003219; 4.657×10^{-4}; 2.178×10^{-4}; 7.324×10^{-5}. R_θ: 3.14×10^{-8}; 2.4×10^{-11}; 2.063×10^{-4}; 0.01049; 0.07453; 0.28011; 0.66591	E-SIM
linked felt geometry	1.5, 2.0, 2.5, 3.0 mm; porosity changes with Eq. (S13)	82.385; 83.020; 83.657; 84.297	97.484; 99.590; 101.494; 103.253	ε_{s}: 0.007188; 0.003219; 0.002065; 9.539×10^{-4}. R_θ: 1.967×10^{-4}; 0.01049; 0.02968; 0.07959	E-SIM
solved flow	25, 50, 100 mL min^{-1}	82.824; 83.020; 83.107	99.425; 99.590; 99.676	ε_{s}: 0.004500; 0.003219; 0.002966. R_θ: 0.01062; 0.01049; 0.01049	E-SIM
salting factor	$\gamma_{\mathrm{salt}} = 2, 3, 4$	108.570; 83.020; 65.162	NR; 99.590; 81.463	ε_{s}: 5.651×10^{-5}; 0.003219; 0.064947. R_θ: 0.72363; 0.01049; 4.71×10^{-8}	E-SIM / E-LIT-PRES
supporting-ligand concentration in the fixed model	2, 3, 4, 5 M	81.112; 82.066; 83.020; 83.973	97.496; 98.541; 99.590; 100.641	ε_{s}: 0.007474; 0.005626; 0.003219; 0.001897. R_θ: 0.005015; 0.007343; 0.01049; 0.01505	E-SIM
liquid conductivity	10, 20, 40 S m^{-1}	83.028; 83.020; 83.013	99.600; 99.590; 99.581	ε_{s}: 0.002996; 0.003219; 0.003353. R_θ: 0.01027; 0.01049; 0.01059	E-SIM
solid conductivity	80, 160, 320 S m^{-1}	83.015; 83.020; 83.026	99.589; 99.590; 99.592	ε_{s}: 0.007193; 0.003219; 0.001598. R_θ: 0.01477; 0.01049; 0.008226	E-SIM
wetted/specific-area factor	0.5, 1.0; no registered converged 2× trajectory	83.017; 83.020	103.316; 99.590	ε_{s}: 5.102×10^{-4}; 0.003219. R_θ: 0.05678; 0.01049	E-SIM
analytical boundary-layer flow	25, 50, 100 mL min^{-1}, $m = 0.5$	74.992; 83.022; 88.733	not evaluated	no solved endpoint state; only the generation contribution is rescaled	E-POST
smooth-permeability substitution	endpoint at $Q = 120$ mAh cm^{-2}	no registered Q_s response	not evaluated	network-minus-baseline endpoint voltage +1.271 mV; $K/K_0 = 0.98250$ versus 0.95629	E-SIM

Among the selected operating and transport variables carried into main Fig. 6, applied current and oxidized-carrier diffusivity produce the largest resolved movements in Q_s. The prescribed $\gamma_{\mathrm{salt}} = 2$–$4$ sensitivity spans a larger total response, but it is an E-LIT/PRES prior rather than a fitted electrolyte law or independent experimental axis. The linked felt and solved-flow families move the marker much less. Conductivity primarily changes the voltage/current- distribution channel, while the saturation marker remains effectively fixed.

S7.2 Analytical sub-grid flow scenario

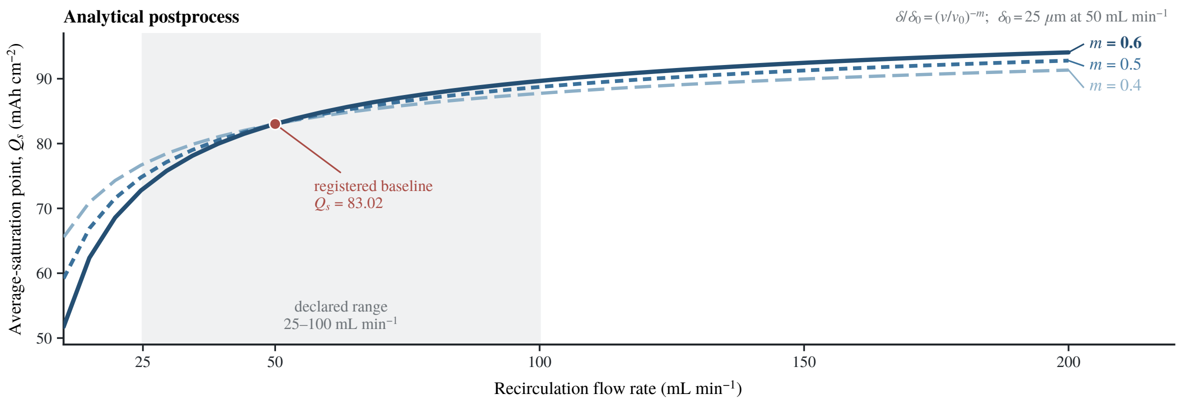

Figure S12: Conditional flow dependence of the analytical boundary-layer scenario. The registered baseline was postprocessed with $\delta/\delta_0 = (v/v_0)^{-m}$ for $m = 0.4$, 0.5 and 0.6, with $\delta_0 = 25\,\mu\mathrm{m}$ at $50\,\mathrm{mL\,min^{-1}}$. All curves pass through $Q_s = 83.020\,\mathrm{mAh\,cm^{-2}}$ at nominal flow. The shaded band marks $25\,\mathrm{mL\,min^{-1}}$ to $100\,\mathrm{mL\,min^{-1}}$; the $m = 0.5$ scenario moves Q_s from 74.992 to $88.733\,\mathrm{mAh\,cm^{-2}}$. These are registered analytical evaluations that rescale only sub-grid generation stress, not a full flow-dependent boundary-layer solve or a measured mass-transfer law.

S7.3 Smooth-permeability substitution

Figure S13 compares the matched smooth-permeability paths and their voltage difference; the calculation does not resolve local pore-throat blockage.

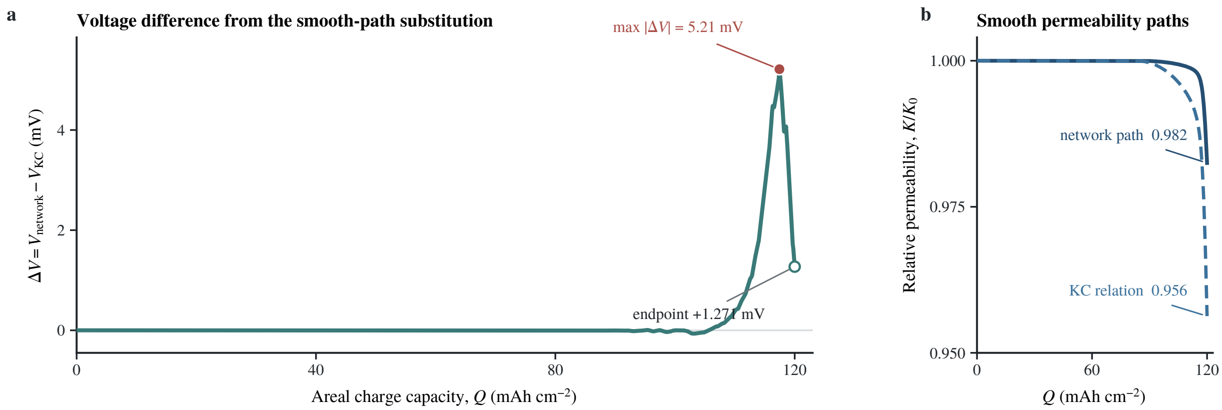

Figure S13: Voltage response to replacing the smooth permeability relation. a, Voltage difference between the re-solve using the pore-network-derived smooth path and the matched Kozeny–Carman baseline. The maximum absolute difference is 5.214 mV at 117.444 mAh cm^{-2}; the difference at 120 mAh cm^{-2} is +1.271 mV. **b,** Corresponding smooth K/K_0 paths; endpoint values are 0.98250 and 0.95629. This control addresses smooth bulk permeability only. Local pore-throat blockage is not resolved or excluded.

S7.4 Transport-channel accounting

Table S17: Transport channels represented, approximated and excluded. Accessibility, diffusive transfer and smooth hydraulic permeability are distinct model channels; local pore blockage is not resolved.

channel	variable or relation	representation	permitted conclusion
calibration-controlled accessibility	$R_\theta = 1 - \theta_{\mathrm{cal}}$	algebraic calibrated relation	changes the kinetic area factor; $Q_{\mathrm{f},cal}$ is defined by $\theta_{\mathrm{cal}} = 0.5$
native reaction area	$A_{\mathrm{bare}}/A_0 = R_\theta T_{\mathrm{pore}}$	product of calibrated and smooth-pore factors	native area export is not identical to R_θ
diffusive transfer	D_{eff}, δ	solved effective diffusion plus prescribed sub-grid length	controls generation stress; absolute values remain prior dependent
smooth hydraulics	$K(\varepsilon_s), \mu, \mathbf{u}, p$	Darcy flow with smooth permeability	supports in-window smooth pressure/velocity response only
charge transport	σ_s, κ_l	solved current-conservation channel	can redistribute current and voltage without moving the saturation marker appreciably
lumped speciation	c_O, c_R, β	one transported oxidized pool plus algebraic partition	no species-resolved polyiodide field or equilibrium validation
local blockage	pore-throat closure, shells, fronts	excluded	no claim about discrete blockage, pore closure or deposit morphology

S8 Scope, evidence boundaries and reproducibility

S8.1 Consolidated claim boundaries

Table S18: Scope and bounds of the positive-electrode analysis. Each row links a data or modeling limit to the maximum supported claim.

topic	class	direct support	maximum claim
full-cell voltage identifiability	E-EXP/E-POST	reduced voltage fit and coefficient trade-off	voltage does not uniquely identify the inventory-to-accessibility relation
accessibility relation	E-SIM/E-CTRL	calibrated R_θ and one matched alternative	model-form sensitivity, not true deposit morphology
native reaction area identity	E-SIM	exported $A_{\mathrm{bare}}/A_0 = R_\theta T_{\mathrm{pore}}$	R_θ alone is not the native area field
threshold definitions	E-SIM/E-COMP	calibrated, island-half and film-connectivity coordinates	definitions are non-interchangeable; none is a measured material threshold
prescribed/literature constants	E-LIT/PRES	documented input values and sensitivity scans	conditional parameter dependence, not present-electrolyte property identification
representative supporting electrolyte	E-EXP/E-LIT/PRES	corrected records and fixed model condition	an experimental/model boundary condition, not the central mechanism or a transferable electrolyte law
sparse experiments	E-EXP	one cell per condition except two at 3 M; one cell per thickness	descriptive within-cell records only; no population optimum or universal law
smooth versus local hydraulics	E-SIM	Darcy field and smooth-permeability control	small smooth in-window penalty does not exclude local blockage
lumped oxidized carrier	E-SIM	one transported c_O state	no separate I_3^-, I_2Br^- or aqueous-I_2 transport claim
electrode-resolved validation	E-EXP	full-cell voltage/capacity anchors	no independent validation of free iodine, retained solid or accessibility
molecular calculations	E-COMP	finite-cell energies and force-field MD	placement/mobility bounds only; no solution equilibrium or nucleation kinetics
comparator geometry	E-COMP	prescribed fibre and network calculations	not imaging or reconstruction of the felt
unexported fields	—	registered spatial table lacks electrical potentials	no electrical-potential map or inference
outside solved domain	—	negative electrode, shuttle, discharge reset and membrane damage not solved	no mechanistic conclusion for these processes

S8.2 Data and code identity

The immutable tagged R582 reproducibility release contains plotted source tables and deterministic renderers for all 19 main and supplementary figures, the exact manuscript and Supplementary Information sources, registered continuum-model input hashes and study/solution/dataset identities, the adopted single-I_2 CP2K input/output identities, and exact topology, parameter, protocol, metadata and log identities for the ten reported GROMACS cases. The operator-confirmed `EXP-META-001` manifest records that every affected legacy `NH4Cl/` `NH4CL` label denotes NH_4Br while preserving raw filenames and hashes. Raw acquisition files, molecular trajectories and original solved `.mph` files are not duplicated in the lightweight repository; their availability boundary and registered identities are documented there. New continuum calculations operate only on copied, separately named and hash-registered models. No repository DOI had been assigned at submission.

Negative-electrode morphology, shuttle, discharge reset and membrane damage lie outside the solved domain. No negative-electrode calculation is displayed. The submission SI also excludes an internal-overpotential decomposition, unrebuildable single-I_2 charge-density artwork, hypothetical electrolyte extrapolation and far-beyond-window mechanical or hydraulic-failure narratives.

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